

When Does Aggregating Explanations Work?

ImageNet Results

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This report consolidates the released **ImageNet** experiment results. Tables are grouped by model, experimental protocol, and aggregation setting. Metric directions and robustness definitions are stated in every table header and caption.

Protocol	Model	Settings
Paper protocol	ResNet-18	NAIVE, IND, NOISE-S, NOISE-K
Paper protocol	ViT-B/16	NAIVE, IND, NOISE-S, NOISE-K

Table 1: NAIVE results for ImageNet with ResNet-18 (Paper protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized. The test split contains $n = 5000$ samples.

Method	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
	F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\downarrow$	R _{$\bar{F}$} ^g $\downarrow$	R _C ^g $\downarrow$	R _{$\bar{C}$} ^g $\downarrow$	R _F ^p $\downarrow$	R _{$\bar{F}$} ^p $\downarrow$	R _C ^p $\downarrow$	R _{$\bar{C}$} ^p $\downarrow$	R _F ^s $\downarrow$	R _{$\bar{F}$} ^s $\downarrow$	R _C ^s $\downarrow$	R _{$\bar{C}$} ^s $\downarrow$	R _F ^a $\downarrow$	R _{$\bar{F}$} ^a $\downarrow$	R _C ^a $\downarrow$	R _{$\bar{C}$} ^a $\downarrow$
Best Individual	0.172	0.175	0.292	0.295	0.078	0.011	0.234	0.184	0.044	0.009	0.204	0.185	0.024	0.000	0.064	0.047	0.036	0.003	0.105	0.002
Simple Averaging	0.173	0.251	0.286	0.386	0.156	0.112	0.347	0.321	0.138	0.092	0.315	0.295	0.073	0.064	0.119	0.120	0.094	0.034	0.177	0.075
Borda	0.158	0.219	0.276	0.346	0.169	0.100	0.338	0.321	0.162	0.079	0.321	0.282	0.075	0.043	0.114	0.103	0.095	0.019	0.188	0.053
Kemeny–Young	0.156	0.223	0.271	0.349	0.167	0.068	0.330	0.277	0.161	0.045	0.317	0.252	0.075	0.036	0.114	0.099	0.090	0.015	0.179	0.049
RRF	0.161	0.218	0.281	0.342	0.165	0.078	0.329	0.294	0.162	0.066	0.313	0.265	0.074	0.034	0.113	0.096	0.096	0.025	0.191	0.061
Schulze	0.158	0.218	0.276	0.344	0.166	0.076	0.330	0.288	0.159	0.066	0.312	0.270	0.077	0.035	0.112	0.096	0.093	0.017	0.186	0.051

Table 2: IND results for **ImageNet** with **ResNet-18** (Paper protocol). The Best Individual row is followed by matched NAIVE and IND results for each aggregation method at $q = 11$. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized. The test split contains $n = 5000$ samples.

Method	Setting	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\downarrow$	R _{$\bar{F}$} ^g $\downarrow$	R _C ^g $\downarrow$	R _{$\bar{C}$} ^g $\downarrow$	R _F ^p $\downarrow$	R _{$\bar{F}$} ^p $\downarrow$	R _C ^p $\downarrow$	R _{$\bar{C}$} ^p $\downarrow$	R _F ^s $\downarrow$	R _{$\bar{F}$} ^s $\downarrow$	R _C ^s $\downarrow$	R _{$\bar{C}$} ^s $\downarrow$	R _F ^a $\downarrow$	R _{$\bar{F}$} ^a $\downarrow$	R _C ^a $\downarrow$	R _{$\bar{C}$} ^a $\downarrow$
Best Individual	baseline	0.044	0.427	0.125	0.569	0.066	0.074	0.238	0.282	0.049	0.072	0.218	0.267	0.025	0.077	0.070	0.132	0.036	0.006	0.109	0.018
Simple Averaging	matched NAIVE	0.027	0.641	0.096	0.795	0.077	0.068	0.260	0.137	0.059	0.067	0.230	0.126	0.023	0.000	0.073	0.054	0.024	0.002	0.089	0.001
Simple Averaging	IND	0.022	0.630	0.091	0.782	0.080	0.064	0.256	0.144	0.058	0.058	0.232	0.139	0.027	0.001	0.077	0.058	0.019	0.001	0.084	0.000
Borda	matched NAIVE	0.031	0.622	0.101	0.776	0.072	0.047	0.256	0.160	0.055	0.041	0.226	0.152	0.026	0.013	0.075	0.070	0.028	0.001	0.093	0.001
Borda	IND	0.029	0.629	0.105	0.782	0.077	0.050	0.251	0.158	0.047	0.042	0.218	0.155	0.025	0.013	0.070	0.069	0.026	0.001	0.099	0.001
Kemeny–Young	matched NAIVE	0.029	0.623	0.097	0.776	0.063	0.048	0.243	0.159	0.047	0.040	0.218	0.153	0.020	0.011	0.066	0.068	0.026	0.003	0.090	0.004
Kemeny–Young	IND	0.030	0.620	0.100	0.776	0.066	0.041	0.247	0.163	0.047	0.037	0.214	0.159	0.022	0.017	0.066	0.071	0.027	0.001	0.093	0.003
RRF	matched NAIVE	0.031	0.624	0.101	0.778	0.068	0.048	0.246	0.159	0.046	0.043	0.219	0.150	0.021	0.012	0.068	0.068	0.029	0.000	0.095	0.002
RRF	IND	0.031	0.625	0.103	0.780	0.066	0.045	0.242	0.162	0.043	0.041	0.213	0.157	0.021	0.016	0.062	0.072	0.028	0.004	0.096	0.006
Schulze	matched NAIVE	0.030	0.622	0.098	0.776	0.064	0.046	0.245	0.160	0.047	0.039	0.219	0.153	0.020	0.012	0.068	0.069	0.027	0.000	0.091	0.002
Schulze	IND	0.030	0.628	0.100	0.781	0.069	0.049	0.248	0.161	0.049	0.044	0.219	0.154	0.022	0.012	0.068	0.071	0.027	0.004	0.094	0.007

Table 3: NOISE-S results for **ImageNet** with **ResNet-18** (Paper protocol), using the Spearman-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\downarrow$	R _F ^g $\downarrow$	R _C ^g $\downarrow$	R _C ^g $\downarrow$	R _F ^p $\downarrow$	R _F ^p $\downarrow$	R _C ^p $\downarrow$	R _C ^p $\downarrow$	R _F ^s $\downarrow$	R _F ^s $\downarrow$	R _C ^s $\downarrow$	R _C ^s $\downarrow$	R _F ^a $\downarrow$	R _F ^a $\downarrow$	R _C ^a $\downarrow$	R _C ^a $\downarrow$
Simple Averaging	4	0.180	0.238	0.302	0.371	0.153	0.118	0.337	0.328	0.139	0.103	0.309	0.310	0.080	0.069	0.121	0.123	0.097	0.034	0.188	0.079
Borda	4	0.186	0.180	0.312	0.302	0.172	0.113	0.334	0.326	0.161	0.093	0.317	0.300	0.080	0.040	0.118	0.102	0.110	0.030	0.206	0.071
Kemeny–Young	4	0.175	0.165	0.298	0.277	0.171	0.086	0.329	0.309	0.161	0.061	0.311	0.276	0.081	0.035	0.114	0.102	0.107	0.009	0.205	0.040
RRF	4	0.183	0.161	0.307	0.275	0.176	0.098	0.333	0.320	0.165	0.072	0.320	0.284	0.083	0.039	0.119	0.102	0.110	0.026	0.205	0.064
Schulze	4	0.170	0.173	0.297	0.287	0.176	0.082	0.324	0.300	0.169	0.051	0.318	0.267	0.088	0.028	0.116	0.092	0.105	0.014	0.206	0.051

Table 4: NOISE-K results for **ImageNet** with **ResNet-18** (Paper protocol), using the Kendall-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\downarrow$	R _F ^g $\downarrow$	R _C ^g $\downarrow$	R _C ^g $\downarrow$	R _F ^p $\downarrow$	R _F ^p $\downarrow$	R _C ^p $\downarrow$	R _C ^p $\downarrow$	R _F ^s $\downarrow$	R _F ^s $\downarrow$	R _C ^s $\downarrow$	R _C ^s $\downarrow$	R _F ^a $\downarrow$	R _F ^a $\downarrow$	R _C ^a $\downarrow$	R _C ^a $\downarrow$
Simple Averaging	5	0.173	0.270	0.287	0.400	0.156	0.126	0.347	0.343	0.129	0.107	0.309	0.317	0.072	0.067	0.119	0.130	0.091	0.043	0.173	0.078
Borda	5	0.188	0.198	0.312	0.322	0.168	0.130	0.341	0.350	0.150	0.111	0.313	0.319	0.076	0.051	0.114	0.114	0.111	0.026	0.203	0.072
Kemeny–Young	5	0.187	0.166	0.313	0.287	0.170	0.096	0.332	0.306	0.163	0.068	0.316	0.277	0.085	0.045	0.120	0.103	0.111	0.027	0.205	0.075
RRF	5	0.189	0.187	0.315	0.309	0.165	0.114	0.335	0.332	0.157	0.097	0.315	0.307	0.082	0.047	0.115	0.108	0.110	0.032	0.205	0.075
Schulze	5	0.188	0.202	0.312	0.326	0.169	0.118	0.337	0.334	0.158	0.100	0.321	0.307	0.079	0.049	0.115	0.111	0.109	0.035	0.202	0.080

Table 5: NAIVE results for ImageNet with ViT-B/16 (Paper protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized. The test split contains $n = 5000$ samples.

Method	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
	F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	$R_F^g \downarrow$	$R_{\bar{F}}^g \downarrow$	$R_C^g \downarrow$	$R_{\bar{C}}^g \downarrow$	$R_F^p \downarrow$	$R_{\bar{F}}^p \downarrow$	$R_C^p \downarrow$	$R_{\bar{C}}^p \downarrow$	$R_F^s \downarrow$	$R_{\bar{F}}^s \downarrow$	$R_C^s \downarrow$	$R_{\bar{C}}^s \downarrow$	$R_F^a \downarrow$	$R_{\bar{F}}^a \downarrow$	$R_C^a \downarrow$	$R_{\bar{C}}^a \downarrow$
Best Individual	0.084	0.095	0.134	0.143	0.011	0.107	0.026	0.133	0.006	0.074	0.023	0.104	0.002	0.013	0.008	0.020	0.022	0.076	0.033	0.077
Simple Averaging	0.064	0.117	0.103	0.174	0.077	0.249	0.110	0.291	0.067	0.181	0.096	0.221	0.024	0.066	0.032	0.081	0.063	0.270	0.079	0.285
Borda	0.048	0.226	0.080	0.287	0.079	0.183	0.113	0.223	0.069	0.123	0.101	0.162	0.027	0.040	0.037	0.054	0.074	0.244	0.095	0.256
Kemeny–Young	0.050	0.268	0.086	0.332	0.085	0.161	0.117	0.199	0.076	0.110	0.104	0.146	0.032	0.038	0.040	0.048	0.071	0.211	0.093	0.217
RRF	0.054	0.217	0.089	0.280	0.083	0.191	0.118	0.230	0.074	0.132	0.104	0.172	0.031	0.052	0.040	0.066	0.070	0.238	0.089	0.243
Schulze	0.049	0.234	0.085	0.297	0.086	0.172	0.119	0.211	0.076	0.122	0.102	0.159	0.034	0.047	0.041	0.056	0.072	0.236	0.090	0.242

Table 6: IND results for ImageNet with ViT-B/16 (Paper protocol). The Best Individual row is followed by matched NAIVE and IND results for each aggregation method at $q = 11$. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized. The test split contains $n = 5000$ samples.

Method	Setting	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\downarrow$	R _{$\bar{F}$} ^g $\downarrow$	R _C ^g $\downarrow$	R _{$\bar{C}$} ^g $\downarrow$	R _F ^p $\downarrow$	R _{$\bar{F}$} ^p $\downarrow$	R _C ^p $\downarrow$	R _{$\bar{C}$} ^p $\downarrow$	R _F ^s $\downarrow$	R _{$\bar{F}$} ^s $\downarrow$	R _C ^s $\downarrow$	R _{$\bar{C}$} ^s $\downarrow$	R _F ^a $\downarrow$	R _{$\bar{F}$} ^a $\downarrow$	R _C ^a $\downarrow$	R _{$\bar{C}$} ^a $\downarrow$
Best Individual	baseline	0.018	0.240	0.038	0.308	0.025	0.269	0.040	0.308	0.015	0.232	0.034	0.267	0.008	0.099	0.012	0.107	0.044	0.104	0.054	0.108
Simple Averaging	matched NAIVE	0.012	0.290	0.029	0.359	0.020	0.298	0.041	0.336	0.009	0.256	0.031	0.293	0.007	0.115	0.015	0.126	0.042	0.102	0.054	0.107
Simple Averaging	IND	0.016	0.303	0.036	0.370	0.019	0.294	0.035	0.332	0.009	0.250	0.027	0.291	0.007	0.100	0.012	0.116	0.041	0.100	0.051	0.107
Borda	matched NAIVE	0.016	0.263	0.033	0.329	0.021	0.289	0.042	0.330	0.012	0.250	0.034	0.290	0.008	0.108	0.017	0.121	0.042	0.097	0.054	0.104
Borda	IND	0.017	0.280	0.038	0.349	0.022	0.285	0.041	0.324	0.013	0.249	0.036	0.287	0.009	0.100	0.014	0.109	0.045	0.103	0.053	0.111
Kemeny–Young	matched NAIVE	0.018	0.253	0.038	0.320	0.024	0.283	0.047	0.322	0.012	0.244	0.037	0.283	0.008	0.104	0.017	0.114	0.044	0.097	0.056	0.101
Kemeny–Young	IND	0.019	0.269	0.039	0.340	0.027	0.283	0.051	0.319	0.014	0.247	0.039	0.284	0.011	0.100	0.019	0.103	0.047	0.100	0.060	0.106
RRF	matched NAIVE	0.017	0.255	0.035	0.322	0.026	0.285	0.048	0.326	0.015	0.243	0.039	0.283	0.009	0.105	0.019	0.115	0.044	0.098	0.057	0.105
RRF	IND	0.018	0.272	0.037	0.338	0.027	0.289	0.052	0.331	0.018	0.244	0.041	0.284	0.013	0.100	0.020	0.110	0.048	0.105	0.058	0.112
Schulze	matched NAIVE	0.019	0.256	0.036	0.323	0.022	0.285	0.046	0.325	0.011	0.247	0.038	0.286	0.008	0.104	0.018	0.114	0.042	0.096	0.055	0.101
Schulze	IND	0.018	0.273	0.036	0.342	0.025	0.291	0.047	0.329	0.015	0.250	0.042	0.289	0.012	0.099	0.019	0.109	0.046	0.103	0.056	0.109

Table 7: NOISE-S results for ImageNet with ViT-B/16 (Paper protocol), using the Spearman-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\downarrow$	R _F ^g $\downarrow$	R _C ^g $\downarrow$	R _C ^g $\downarrow$	R _F ^p $\downarrow$	R _F ^p $\downarrow$	R _C ^p $\downarrow$	R _C ^p $\downarrow$	R _F ^s $\downarrow$	R _F ^s $\downarrow$	R _C ^s $\downarrow$	R _C ^s $\downarrow$	R _F ^a $\downarrow$	R _F ^a $\downarrow$	R _C ^a $\downarrow$	R _C ^a $\downarrow$
Simple Averaging	2	0.085	0.097	0.132	0.148	0.067	0.220	0.083	0.272	0.055	0.171	0.071	0.219	0.023	0.064	0.021	0.082	0.077	0.198	0.093	0.205
Borda	2	0.086	0.093	0.135	0.146	0.073	0.223	0.085	0.272	0.054	0.168	0.068	0.214	0.029	0.062	0.025	0.081	0.076	0.203	0.091	0.210
Kemeny–Young	2	0.086	0.093	0.135	0.146	0.073	0.223	0.085	0.272	0.054	0.168	0.068	0.214	0.029	0.062	0.025	0.081	0.076	0.203	0.091	0.210
RRF	2	0.087	0.092	0.136	0.144	0.074	0.225	0.086	0.274	0.053	0.169	0.068	0.216	0.026	0.064	0.022	0.083	0.076	0.204	0.090	0.210
Schulze	2	0.086	0.093	0.135	0.146	0.073	0.223	0.085	0.272	0.054	0.168	0.068	0.214	0.029	0.062	0.025	0.081	0.076	0.203	0.091	0.210

Table 8: NOISE-K results for **ImageNet** with ViT-B/16 (Paper protocol), using the Kendall-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is the absolute quality change under perturbation; lower values are better and all R columns are minimized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\downarrow$	R _F ^g $\downarrow$	R _C ^g $\downarrow$	R _C ^g $\downarrow$	R _F ^p $\downarrow$	R _F ^p $\downarrow$	R _C ^p $\downarrow$	R _C ^p $\downarrow$	R _F ^s $\downarrow$	R _F ^s $\downarrow$	R _C ^s $\downarrow$	R _C ^s $\downarrow$	R _F ^a $\downarrow$	R _F ^a $\downarrow$	R _C ^a $\downarrow$	R _C ^a $\downarrow$
Simple Averaging	3	0.073	0.097	0.120	0.148	0.083	0.232	0.108	0.280	0.068	0.174	0.094	0.220	0.031	0.057	0.032	0.076	0.075	0.199	0.093	0.204
Borda	3	0.051	0.103	0.096	0.151	0.080	0.252	0.102	0.303	0.064	0.187	0.090	0.235	0.024	0.067	0.024	0.089	0.102	0.191	0.121	0.201
Kemeny–Young	3	0.087	0.095	0.136	0.146	0.072	0.220	0.091	0.271	0.058	0.168	0.079	0.214	0.027	0.059	0.027	0.077	0.077	0.207	0.093	0.216
RRF	3	0.073	0.096	0.123	0.147	0.082	0.224	0.103	0.273	0.068	0.156	0.086	0.205	0.031	0.049	0.028	0.071	0.086	0.202	0.102	0.206
Schulze	3	0.076	0.098	0.125	0.148	0.074	0.236	0.096	0.286	0.062	0.175	0.083	0.224	0.023	0.061	0.023	0.079	0.079	0.191	0.097	0.199